



Lagrangian algorithm with steepest descent method applied to the planar mechanism synthesis problem using natural coordinates

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Abstract This research introduces an approach, denoted as the Natural Coordinates Lagrangian Algorithm using the Steepest Descent Method ($NCLA_{sdm}$) for the dimensional synthesis of planar mechanisms with the objective of determining the required geometric dimensions for the generation of a desired path. The proposed approach utilizes a Lagrangian optimization method that employs the Steepest Descent Method and natural coordinates related by the constraint equations defining the admissible configurations of the mechanism. The proposed method is validated through the synthesis of four mechanisms of varying complexity. The numerical examples shown are slider-crank mechanisms with and without slider eccentricity, a four-bar mechanism, and the Theo Jansen linkage. The results demonstrate a strong agreement between the desired paths and those generated by the synthesized mechanisms,

even in highly complex mechanisms such as those of Theo Jansen.

Keywords Dimensional synthesis · Path generation · Lagrangian method · Four-bar linkage · Slider-crank linkage · Jansen linkage

List of symbols

IC	Initial condition
NC	Natural coordinates
n	Number of natural coordinates
m	Number of constraint equations
n_b	Number of bars in the mechanism
n_d	Number of points in the desired path
l_i	Length of the bar i of the mechanism
a_i	Geometrical parameter i of the mechanism
q	Vector of dependent parameters for the generated path.
q'	Vector of coordinates that define the desired path.
q_i	Vector of the i independent coordinates.
q_j	Vector of the j dependent coordinates.
q'_j	Vector of the j dependent coordinates of the points in the desired path.
α	Input parameter or independent parameter

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λ_j	Lagrange multiplier corresponding to j dependent coordinates
$L(l_i, q_j, \lambda_j)$	Lagrangian function dependent on l_i , q_j and λ_j .
$J(l_i)$	Error function or objective function
$\phi(q_j)$	Vector of constraint equations or function vector.
ϕ_j	Constraint equations that define the mechanism.
ϕ_q	Jacobian matrix of dependent terms of the constraint equations
ϕ_{l_i}	Matrix of the derivatives of the constraint equations with respect to l_i
p_z	Search direction
h	Step size
h_0	Initial step size
h^*	Step size that minimizes the function $\varphi(h)$.
k	Iterator associated with the number of points n_d of the desired path.

1 Introduction

In the conceptual design phase, designers explore various mechanism types to determine which one aligns best with the design goals. Consequently, during this phase, mechanism synthesis is performed and is typically divided into type, number, and dimensional synthesis. The authors of Ye and Li (2019) present a comprehensive review of the different types of synthesis methods based on different theories, categorizing the approaches outlined in the literature into three main groups: motion-based methods, constraint-based methods, and alternative methods.

The planar mechanism synthesis problem presents a significant challenge for mechanical engineers because it involves designing a mechanism to perform a specific task. This synthesis problem encompasses function generation, path generation, and rigid-body guidance. In all cases, it is necessary to define the desired points or poses that the tracking point of a mechanism must pass through or attain. Dimensional synthesis, for a given mechanism, aims to determine the dimensions of its links to achieve the desired task.

A review of advances in the dimensional synthesis of four-bar mechanisms from the 1970s to the present day is presented in Lee and Russell (2018). Synthesis methodologies are mainly classified into two fam-

ilies: heuristic methods and gradient-based methods. Heuristic methods proposed in Ebrahimi and Payvandy (2015) employ an Imperialist Competitive Algorithm (ICA) to minimize the tracking error of the mechanism coupler point with respect to a desired path. Including workspace bounds and Grashof's law as a seed to generate an acceptable initial population improves the optimization performance. An Artificial Neural Network (ANN) with Fourier descriptors for approximate dimensional synthesis of planar linkages is proposed in Khan et al. (2015). The method's success, illustrated by a four-bar linkage generating a simple closed trajectory, depends on how closely the generated path approximates the desired path. A detailed optimal dimensional synthesis of a six-bar steering mechanism is presented by Romero-Núñez and Tuiran-Villalba (2018). A Genetic Algorithm (GA) with Natural Coordinates (NC) for the kinematic model is used to derive optimal mechanism dimensions tailored to a specific vehicle's geometry. The use of a hybrid Genetic Algorithm-Fuzzy Logic (GA-FL) approach is shown to outperform stand-alone genetic algorithms in performance for dimensional synthesis of four-bar mechanisms, as illustrated in a closed trajectory generation task Laribi et al. (2004). IOA^{s-ot} is a self-adapting optimization algorithm based on Differential Evolution (DE) that uniquely employs dynamic rather than constant control parameters Ortiz et al. (2013). This methodology is proposed for the dimensional synthesis of path-generating linkages, and its efficacy is demonstrated through application to both four- and six-bar mechanisms. A metaheuristic technique, the Cuckoo Search Algorithm, is used in Bulatovic et al. (2013) for the dimensional synthesis of double-dwell six-bar mechanisms, with a specific application to the Stephenson III linkage. A modified Krill Herd algorithm for optimal dimensional synthesis is proposed in Bulatović et al. (2016); it is demonstrated through path generation for a four-bar mechanism.

Gradient-based methods are widely used in mechanism synthesis, with early approaches minimizing loop closure equation (LCE) errors to define dimensions Hernández et al. (2021). These are specific cases of optimization techniques that minimize the Euclidean or synthesis error function (EEF/SEF), defined as the distance between the generated and desired paths. Their relative complexity remains a key consideration. Heuristic methods are generally more complex than gradient-based approaches Hernández et al. (2021). In

Mariappan and Krishnamurty (1996), an Exact Gradient Method (EGM) employing matrix notation is presented to minimize the SEF, with mechanism configurations derived from Loop Closure Equations (LCE). The method can be applied to many classes of mechanism synthesis problems. Its effectiveness is demonstrated through the dimensional synthesis of a six-bar punching mechanism and an eight-bar dwell mechanism. An EGM for path synthesis in planar mechanisms with numerous precision points is presented in Sancibrian et al. (2004). The proposed optimal synthesis approach is based on sensitivity analysis and uses exact derivatives to minimize the SEF. The method features low computational cost and accommodates several kinds of constraints, including link lengths, overall linkage dimensions, and mobility ranges. Its effectiveness is illustrated through three numerical examples involving four-bar and six-bar mechanisms. A gradient method incorporates the slope of the tangent at each point of the path into the SEF, as shown in Hernández et al. (2021). The method analyzes all partial derivatives analytically and is applied to four-bar mechanisms following complex paths. Reference Jiménez et al. (1997) details an optimal multi-body kinematic synthesis method employing NC for the equations defining the mechanism configuration. The authors address the potential nonexistence of solutions for an objective function that combines constraint equations and the SEF by reducing the problem to minimizing the SEF alone. Applications to planar and 3D mechanisms are presented. Reference Avilés et al. (2010) details an enhanced, general method for optimal linkage synthesis using finite element models and an energy-based error function. Optimal rod lengths (optimization parameters) are found by minimizing the model energy subject to synthesis constraints. Error is evaluated by determining elastic equilibrium at precision points via Hessian minimization, with Lagrange multipliers to handle rod non-deformability constraints. This dimensional synthesis technique applies to planar linkages with revolute and prismatic joints. Reference Da Lio et al. (2000) presents an NC-based synthesis method in which constraints and requirements are satisfied by identifying the closest candidate mechanism. This generates an initial value problem for a system of ordinary differential equations, thereby avoiding the need to assemble the mechanism in its initial configuration according to the initial conditions. The authors De Jalon and Bayo (2012) show that the use of the

NC enables a simple representation of multibody systems and yields important advantages for kinematic and dynamic simulation. Additional researchers, such as Li and Wang (2014), advocate an approach that employs NC to address rigid multibody dynamics. This method not only retains the benefits of traditional formulations but also enhances the efficiency of constraint-equation modeling, offering significant advantages for automated modeling.

This study introduces the $NCLA_{sdm}$ method, a dimensional synthesis framework for planar mechanisms designed to determine the geometric dimensions required to reproduce a prescribed path through an Lagrangian optimization approach. The method employs NC parameterization that eliminates the need for Loop Closure Equations involving transcendental functions and angular variables characteristic of conventional formulations. While the parameters used to define the mechanism's configuration are referred to as NC, following established multibody dynamics nomenclature De Jalon and Bayo (2012), they can also be viewed as mixed coordinates depending on the specific literature context. All kinematic relations are expressed in NC, yielding predominantly quadratic or linear constraint equations that enable exact analytical gradient computation, improved numerical stability, and higher computational efficiency. The Lagrangian formulation unifies bar lengths, dependent coordinates, and Lagrange multipliers into a single constrained optimization structure, ensuring continuous kinematic admissibility and mechanism assemblability. A tailored Steepest Descent strategy (in which the optimal step size h^* is obtained via a Taylor expansion and iteratively refined using the Newton–Raphson procedure) enhances both directional stability and convergence robustness. Compared with heuristic algorithms such as Genetic Algorithms, Differential Evolution, and the Imperialist Competitive Algorithm, $NCLA_{sdm}$ employs a deterministic scheme that exploits intrinsic kinematic sensitivities to achieve faster, more coherent convergence at a considerably lower computational cost. Unlike stochastic black-box searches that require extensive function evaluations and empirical parameter tuning, the proposed method ensures physically consistent synthesis by explicitly incorporating kinematic constraints and provides a predictable first-order convergence path while maintaining stability near local minima. Its algorithmic simplicity, achieved without resorting to higher-order derivatives or matrix-

intensive operations, is demonstrated through applications to mechanisms of varying complexity, including a slider-crank, a four-bar, an offset sliding-crank, and the Theo Jansen linkage.

2 Lagrangian method for the mechanism synthesis problem

The main procedures used for the dimensional synthesis of mechanisms involve minimizing an objective function. This function is the difference between the path generated by a point in the mechanism and the desired path. Thus, the optimization problem consists of minimizing an error function $J(l_i)$ defined by squaring the Euclidean norm between the path generated by a point of the mechanism and the desired path, as shown in Eq. (1).

$$J(l_i) = \text{Min}[\|\mathbf{q} - \mathbf{q}'\|^2] \quad (1)$$

where $\mathbf{q}$ is the vector of the dependent parameters of the mechanism that have been defined using NC and $\mathbf{q}'$ is the vector of coordinates that define the theoretical or desired path.

The proposed optimization method starts from the Lagrangian function, which is a multivariate function Eq. (2) that depends on the lengths of the bars l_i , the dependent coordinates q_j and the Lagrange multipliers λ_j Nocedal and Wright (2006).

$$L(l_i, q_j, \lambda_j) = \sum_{k=1}^{n_d} \left(\|\mathbf{q}_j - \mathbf{q}'_j\|^2 + \lambda_j^T \boldsymbol{\phi}(q_j) \right)_k \quad (2)$$

where $\boldsymbol{\phi}(q_j)$ is a vector of constraint equations containing the ϕ_j constraint equations that define the mechanism and n_d is the number of points defining the desired path.

The minimization of the Lagrangian Eq. (2) is equivalent to minimizing the error function Eq. (1). The minimum of the Lagrangian function is found by differentiating the function with respect to the variables of interest. In this case, the variables are the lengths l_i of the bars of a mechanism, which in turn depend on the NC of the system and the Lagrange multipliers.

$$\frac{dL}{dl_i} = \frac{\partial L}{\partial l_i} + \frac{\partial L}{\partial q_j} \frac{\partial q_j}{\partial l_i} + \frac{\partial L}{\partial \lambda_j} \frac{\partial \lambda_j}{\partial l_i} \quad (3)$$

The partial derivative of the Lagrangian function L with respect to the Lagrangian multiplier λ_j Eq. (4) takes the following form:

$$\frac{\partial L}{\partial \lambda_j} = \sum_{k=1}^{n_d} [\phi_j(q_j)]_k = 0 \quad (4)$$

Expanding Eq. (4), the partial derivative $\partial L / \partial \lambda_j$ is equal to the function vector $\boldsymbol{\phi}(q_j)$, which by definition is zero since it is the system formed by the constraint equations defined in NC. Accordingly, the partial derivative of the Lagrangian function L with respect to the vector q_j is:

$$\frac{\partial L}{\partial q_j} = \sum_{k=1}^{n_d} \left[2(q_j - q'_j) + \lambda_j^T \boldsymbol{\phi}_q \right]_k \quad (5)$$

where $\boldsymbol{\phi}_q$ is the Jacobian matrix of the dependent terms. The optimum of the function is found by differentiating and setting it equal to zero. The k terms in Eq. (5) must be zero, thus obtaining the j Lagrange multipliers λ_j . Rearranging Eq. (5), the Lagrange multipliers are obtained by solving Eq. (6) below.

$$[\boldsymbol{\phi}_q^T]_k \lambda_{j,k} = -2(q_j - q'_j)_k \quad (6)$$

Continuing with the differential terms of Eq. (3), the partial derivative of the Lagrangian function L with respect to the lengths of the bars l_i , is expressed as:

$$\frac{\partial L}{\partial l_i} = \sum_{k=1}^{n_d} \left(\lambda_j^T \frac{\partial \phi_j}{\partial l_i} \right)_k \quad (7)$$

where $\partial \phi_j / \partial l_i$ is the partial derivative of the constraint equations with respect to the length of the bars l_i .

Analyzing the terms of Eq. (3), it follows that the derivative of the error function in Eq. (1) is equal to the partial derivative of the Lagrangian function in Eq. (2) with respect to the lengths of the bars, provided that equations Eq. (4) and Eq. (6) are satisfied. As described above, Eq. (4) is satisfied by definition, whereas Eq. (6) must be satisfied. Substituting the Lagrange multipliers obtained from Eq. (6) into Eq. (7) satisfies the equation below.

$$\frac{dJ}{dl_i} = \frac{\partial L}{\partial l_i} \quad (8)$$

With this last Eq. (8), there is an estimator of the slope or derivative of the objective function J for a given bar length l_i .

2.1 Iterative method and pseudocode

The steepest descent method (SDM) is the most fundamental gradient-based optimization approach for n variables. While second-order methods, such as Newton–Raphson, typically offer faster convergence, they are highly sensitive to initial guesses and frequently fail to converge in complex, non-linear kinematic syntheses Bartholomew-Biggs (2008). In this study, SDM was preferred for its superior robustness to the non-linearities inherent in closed-loop mechanisms. Throughout the numerical examples presented, no significant convergence failures were observed, demonstrating that the method reliably reaches the neighborhood of the local minimum for the evaluated topologies.

The implementation of $NCLA_{sdm}$ follows the standard algorithmic structure described in Bartholomew-Biggs (2008). As shown in the pseudo-code below, each iteration consists of two critical phases: the determination of a search direction, p_z , followed by a line search to identify a suitable step size, h^* . In this model, the step size is considered exact when it effectively minimizes the objective function $\varphi(h)$.

Choose an initial estimate, $l_{i,0}$, for the minimum of $L(l_{i,z}, q_j, \lambda_j)$.
 Repeat for $z = 0, 1, 2, \dots$
 set

$$p_z = \frac{\partial L(l_{i,z}, q_j, \lambda_j)}{\partial l_{i,z}}$$

 calculate h^* to minimize
 $\varphi(h) = L(l_{i,z} + h \cdot p_z, q_j, \lambda_j)$
 set $l_{i,z+1} = l_{i,z} + h \cdot p_z$
 until $\|\delta L_{z+1}\|$ is sufficiently small.

The derivative of $\varphi(h)$ can be approximated using a Taylor series Bartholomew-Biggs (2008). The solution allows the appropriate step size to be found, as shown in the following equation.

$$\sum_{k=1}^{N_p} p_z^T \frac{\partial L_k(l_i + h \cdot p_z, q_j, \lambda_j)}{\partial l_i} = 0 \quad (9)$$

This work applies the iterative Newton–Raphson method to find an approximate solution h^* of Eq. (9), starting from an initial value h_0 . The initial value is determined empirically (trial-and-error approach). If the initial value is very far from the solution, the steepest descent method diverges. For this purpose, two or three iterations are defined, and it is checked that the steepest descent method converges for the initial value h_0 taken. If this is not the case, the h_0 value is replaced by a smaller one and the convergence is checked again. In all cases studied here, we start with $h_0 = 0.1$ and, if necessary, use the smallest value as the initial value for the Newton–Raphson method.

3 Lagrangian Algorithm with steepest descent ($NCLA_{sdm}$) in planar mechanism synthesis

Four examples of applying the method described in this article are provided below. In the first example, the dimensions of the two bars that make up the slider-crank mechanism are determined, and the method is demonstrated manually. The other three examples correspond to the synthesis of complex mechanisms, with multiple geometrical dimensions to be found and more than one possible dimensional configuration as solutions to the same mechanism synthesis problem. The ability of $NCLA_{sdm}$ to converge to these distinct valid configurations as the initial conditions vary is a significant practical advantage. It allows designers to select the most suitable solution based on additional criteria, such as space constraints, without incurring the computational overhead associated with global evolutionary search algorithms. In the following numerical examples, all linear dimensions and coordinates are expressed in length units (e.g., mm, cm or m), and angular parameters are in radians. The input parameter constraint is applied in the dimensional synthesis problems presented here.

3.1 $NCLA_{sdm}$ applied to an inline slider-crank mechanism to generate a desired slider path

The following is a step-by-step example demonstrating the dimensional synthesis method described in this work, applied to one of the most widely used mechanisms, the slider-crank mechanism. The optimization process begins with a set of prescribed slider positions

and determines the optimal link lengths required for the mechanism to track a trajectory that minimizes the error between the actual and desired paths. Figure 1 shows the aforementioned mechanism, the chosen reference system, and the proposed NC to describe its configuration. In this example, the slider has zero eccentricity. The NC vector is $\{\theta, x_2, y_2, x_3\}^T$, being the total number of NC n equals four. It is a mechanism with one degree of freedom (*dof*), and therefore, with only one independent coordinate, q_i equals θ , and three dependent coordinates q_j , these are $\{x_2, y_2, x_3\}$. The number m of constraint equations is the difference between the total number of NC and the *dof*. In this particular case, m equals 3. The objective is to determine the dimensions of the bar lengths a_1 and a_2 so that point 3 satisfies the design constraints: $x'_3(\theta = 0)$ equals 10 and $x'_3(\theta = \pi \text{ rad})$ equals 2. The solution to this problem is known. The lengths of bars a_1 and a_2 are 4 and 6, respectively.

Next, it is shown how the iterative algorithm works to find the bar lengths that allow generating a path as close as possible to the desired path, starting from known initial bar lengths established by the designer through the vector $\{l_{1,0}, l_{2,0} \dots l_{n,0}\}^T$. These initial values must be chosen to be a good approximation to the solution of the problem. The set of constraint equations in NC that define the position problem for this mechanism is contained in the following equation vector $\phi(q_j)$, which by definition is equal to zero De Jalon and Bayo (2012).

$$\phi(q_j) \equiv \begin{Bmatrix} x_2 - a_1 \cos(\theta) \\ y_2 - a_1 \sin(\theta) \\ (x_3 - x_2)^2 + (0 - y_2)^2 - a_2^2 \end{Bmatrix} = 0 \quad (10)$$

Let $l_{1,0}$ and $l_{2,0}$ be a first approximation of the bar lengths a_1 and a_2 respectively. Knowing these lengths, the system of equations Eq. (10) is solved, and the dependent coordinates x_2, y_2, x_3 are found for a given input of the problem. Table 1 shows the solution of the system Eq. (10) for the inputs: k equals 1 where θ equals zero and k equals 2 where θ equals π .

For the mechanism shown in Fig. 1, Eq. (6) for k equals to 1 and 2 takes the form:

$$\begin{bmatrix} 1 & 0 & -2(x_3 - x_2) \\ 0 & 1 & 2y_2 \\ 0 & 0 & 2(x_3 - x_2) \end{bmatrix}_k \begin{Bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{Bmatrix}_k = -2 \begin{Bmatrix} 0 \\ 0 \\ x_3 - x'_3 \end{Bmatrix}_k \quad (11)$$

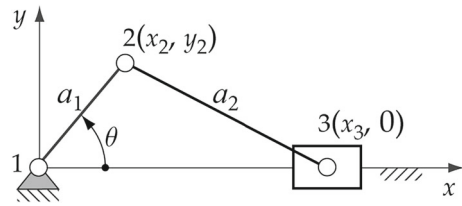


Fig. 1 Kinematic diagram of the inline slider-crank mechanism and the NC used to define its configuration

Table 1 Solutions of the system of equations Eq. (10)

k	θ	x_2	y_2	x_3
1	0	a_1	0	$a_1 + a_2$
2	π	$-a_1$	0	$a_2 - a_1$

Equation (11) for k equals 1 and 2 takes the following form:

$$\begin{bmatrix} 1 & 0 & -2a_2 \\ 0 & 1 & 0 \\ 0 & 0 & 2a_2 \end{bmatrix}_1 \begin{Bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{Bmatrix}_1 = \begin{Bmatrix} 0 \\ 0 \\ a_1 + a_2 - 10 \end{Bmatrix}_1 \quad (11a)$$

$$\begin{bmatrix} 1 & 0 & -2a_2 \\ 0 & 1 & 0 \\ 0 & 0 & 2a_2 \end{bmatrix}_2 \begin{Bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{Bmatrix}_2 = \begin{Bmatrix} 0 \\ 0 \\ a_2 - a_1 - 2 \end{Bmatrix}_2 \quad (11b)$$

Equation (7) for the mechanism under study takes the form:

$$\frac{\partial L}{\partial a_1} = \sum_{k=1}^2 \left((\lambda_1, \lambda_2, \lambda_3)_k \begin{Bmatrix} -\cos \theta \\ -\sin \theta \\ 0 \end{Bmatrix}_k \right) \quad (12a)$$

$$\frac{\partial L}{\partial a_2} = \sum_{k=1}^2 \left((\lambda_1, \lambda_2, \lambda_3)_k \begin{Bmatrix} 0 \\ 0 \\ -2a_2 \end{Bmatrix}_k \right) \quad (12b)$$

Replacing Eq. (11a) and Eq. (11b) in equations Eq. (12a) and Eq. (12b) respectively, the following results:

$$\frac{\partial L}{\partial a_1} = 2(2a_1 - 8) \quad (13a)$$

$$\frac{\partial L}{\partial a_2} = 2(2a_1 - 12) \quad (13b)$$

Because of the simplicity of this example, the step size h can be determined analytically. The step size that maximizes the reduction along the search direction is

Table 2 Solution of the iterations for the bar lengths a_i in a slider-crank mechanism

Iter. k	Bar lengths		$k=1, \theta=0, x'_3=10$			$k=2, \theta=\pi, x'_3=2$			Part. deriv.		Step h
	a_1	a_2	λ_1	λ_2	λ_3	λ_1	λ_2	λ_3	$\partial L/\partial a_1$	$\partial L/\partial a_2$	
1	5.00	5.00	0.00	0.00	0.00	4.00	0.00	0.40	4.00	-4.00	0.25
2	4.00	6.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

shown in the following equation:

$$h = \frac{(a_1 - 4)^2 + (a_2 - 6)^2}{2[(a_1 - 4)(2a_1 - 8) + (a_2 - 6)(2a_2 - 12)]} \quad (14)$$

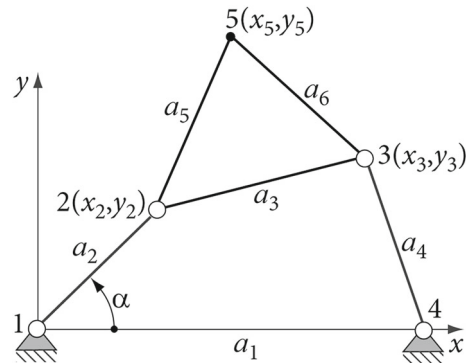
Starting with an initial approximation of link lengths $l_{1,0} = 5$ and $l_{2,0} = 5$, the optimization results are detailed in Table 2. The data in this table demonstrates that two iterations are sufficient to determine the bar dimensions that minimize the error between the desired and generated paths. This rapid convergence validates the efficacy of the proposed dimensional synthesis method.

3.2 $NCLA_{sdm}$ applied to a Grashof four-bar mechanism to generate a desired path

A four-bar mechanism is a mechanism commonly used to test different synthesis algorithms, as shown in (Laribi et al. 2004; Bulatović et al. 2016; Hernández et al. 2021; Smaili and Diab 2007; Cabrera et al. 2011). In the comparative results for Case 2, studied by the authors Ortiz et al. (2013), they use the results of the dimensional synthesis obtained by the authors Kunjur and Krishnamurty, using the synthesis method called K-K Algorithm Kunjur and Krishnamurty (1997), among other methods.

In the second example presented here, the kinematic diagram and NC for the Grashof four-bar mechanism are shown in Fig. 2. The coordinate vector is formed by the parameters $x_2, y_2, x_3, y_3, x_5, y_5$ and α , so a total of seven NC are used, where α is the independent coordinate and the rest are the dependent coordinates. The number of constraint equations is equal to six. The geometrical parameters are a_1, a_2, a_3, a_4, a_5 and a_6 . Table 3 shows the values of the points of the desired path used in this example, which have been taken from Case 2 Ortiz et al. (2013).

The constraint equations of this mechanism are the system formed by the following vector equations.

**Fig. 2** NC and geometrical parameters of the Grashof four-bar mechanism**Table 3** Data points of the desired path Ortiz et al. (2013)

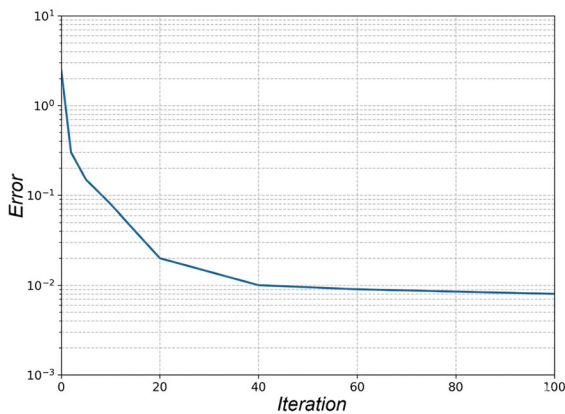
Points j	x_5	y_5	α
1	3.00	3.00	$\pi/6$
2	2.759	3.363	$\pi/4$
3	2.372	3.663	$\pi/3$
4	1.890	3.862	$5\pi/12$
5	1.355	3.943	$5\pi/2$

$$\phi(q_j) \equiv \begin{Bmatrix} x_2 - a_2 \cos(\alpha) \\ y_2 - a_2 \sin(\alpha) \\ (x_3 - x_2)^2 + (y_3 - y_2)^2 - a_3^2 \\ (x_5 - x_2)^2 + (y_5 - y_2)^2 - a_5^2 \\ (x_3 - x_5)^2 + (y_3 - y_5)^2 - a_6^2 \\ (a_1 - x_3)^2 + (y_3)^2 - a_4^2 \end{Bmatrix} = 0 \quad (15)$$

Table 4 shows a comparison between the results obtained by the K-K algorithm Kunjur and Krishnamurty (1997), the IOA^{s-ot} algorithm Ortiz et al. (2013), and the algorithm $NCLA_{sdm}$ proposed in this article. The solution proposed in this article is based on one initial condition (IC). The IC defines dimensions such that the four-bar mechanism meets the Grashof criterion. The synthesis solution is obtained with 500 and 8000 iterations, yielding errors of 3.64×10^{-3} and

Table 4 Four-bar synthesis results according to the K-K, IOA^{s-ot} and $NCLA_{sdm}$ algorithms

Iterations		Synthesis methods		$NCLA_{sdm}$ -First solution		
		K-K Algorithm	IOA^{s-ot}	IC	$h_0 = 0.1$	$h_0 = 0.1$
		5000	5000		500	8000
Parameters a_i	a_1	3.509643	2.8036	4.0	3.6322	3.3213
	a_2	1.8576062	1.99226	2.0	2.3284	2.0037
	a_3	4.725835	3.030461	3.0	3.3783	3.5737
	a_4	3.518721	2.474117	3.0	2.7836	2.5788
	a_5			2.0	2.0681	2.3649
	a_6			2.0	2.3248	2.5698
Error		9.43×10^{-4}	4.47×10^{-6}	2.32	3.64×10^{-3}	2.73×10^{-6}

**Fig. 3** Convergence of error values versus iteration count for the $NCLA_{sdm}$ mechanism synthesis (based on initial conditions from Table 4)

2.73×10^{-6} , respectively. The lengths of bars obtained with the proposed method are very similar to those obtained by the K-K and IOA^{s-ot} algorithms. The convergence error for the $NCLA_{sdm}$ is illustrated in Fig. 3, using the initial conditions specified in Table 4. The curve reveals two distinct convergence phases: a rapid initial decrease in error during the first 20 iterations, followed by a significantly reduced convergence rate as the algorithm approaches the optimal solution. This behavior is characteristic of the steepest descent component, which typically exhibits fast initial progress but slows down near the local minimum.

Six different solutions obtained from different initial conditions are shown in Table 5. All the ICs chosen satisfy the Grashof criterion. All simulations were done with 8000 iterations. While $h_0 = 0.1$ was initially targeted as the step size, smaller values (down to 0.0005)

were implemented in specific cases to ensure algorithmic convergence, as detailed in the table. Despite these variations, nearly all solutions achieved an error of order 10^{-6} .

Table 5 illustrates that the $NCLA_{sdm}$ algorithm maintains high stability and precision across different ICs. Regardless of the starting parameter values, all six synthesis solutions converge to significantly lower error levels, typically in the range of 10^{-5} to 10^{-6} after 8,000 iterations. While individual parameters a_1 through a_6 fluctuate based on the specific IC, the final synthesized dimensions remain physically consistent, demonstrating the algorithm's robustness against the initial value problem common in kinematic synthesis.

The performance of the proposed $NCLA_{sdm}$ was evaluated against established literature benchmarks, specifically the K-K Genetic Algorithm Kunjur and Krishnamurty (1997) and modern IOA^{s-ot} Ortiz et al. (2013) strategies. The K-K method, a deterministic gradient-based optimizer using the Generalized Reduced Gradient algorithm, employs well-defined initial guesses for design variables (crank angle and link lengths constrained by Grashof's law) and converges by minimizing a structural error function as the least-squares Euclidean distance between generated and target precision points, conditions aligned with our formulation and error function $J(li)$. Similarly, the IOA^{s-ot} method, a population-based meta-heuristic derived from Differential Evolution, initializes link lengths randomly within (0,1) while sharing our kinematic parameter set, and converges by minimizing the same Euclidean-norm-based structural error or reaching a maximum iteration limit. In contrast, $NCLA_{sdm}$ achieves a convergence depth of 1.08×10^{-5} within

Table 5 Synthesis solutions of the Grashof four-bar mechanism taking different IC in the $NCLA_{sdm}$

Solutions		First		Second		Third	
Iterations		IC	$\frac{h_0 = 0.1}{8000}$	IC	$\frac{h_0 = 0.01}{8000}$	IC	$\frac{h_0 = 0.0005}{8000}$
Parameters a_i	a_1	4.0	3.321	4.0	3.535	4.0	3.085
	a_2	2.0	2.004	2.0	2.100	2.0	2.005
	a_3	3.0	3.574	3.0	3.788	3.0	3.305
	a_4	3.0	2.579	3.0	2.615	3.0	2.514
	a_5	2.0	2.365	3.0	2.359	1.0	2.364
	a_6	2.0	2.570	3.0	2.734	1.0	2.389
Error		2.32	1.08×10^{-5}	12.90	8.09×10^{-6}	8.34	3.96×10^{-6}
Solutions		Fourth		Fifth		Sixth	
Iterations		IC	$\frac{h_0 = 0.01}{8000}$	IC	$\frac{h_0 = 0.01}{8000}$	IC	$\frac{h_0 = 0.01}{8000}$
Parameters a_i	a_1	2.0	1.808	6.0	4.810	3.0	2.919
	a_2	1.0	1.987	4.0	2.014	3.0	2.001
	a_3	1.5	1.933	4.0	5.320	3.0	3.133
	a_4	1.5	2.251	4.0	3.162	3.0	2.483
	a_5	2.0	2.380	3.0	2.355	2.0	2.367
	a_6	2.0	1.785	3.0	3.930	2.0	2.276
Error		24.3	1.08×10^{-5}	31.3	2.86×10^{-6}	2.89	2.86×10^{-6}

62.9 min, exceeding error floors of standard heuristic and gradient-based methods while overcoming the stability-precision trade-offs in K-K and IOA^{s-ot} through its smoothing-damping architecture.

Figure 4 shows 3 of the solutions obtained in Table 5, which are the first, the fifth and the sixth. It can be seen how the dimensions of the mechanism change. The path followed by point 5 (Fig. 2) of the mechanism also changes, except at the desired points.

3.3 Offset slider-crank mechanism to generate a desired path

A modification to the function generation synthesis methodology is applied to the dimensional synthesis of an offset slider-crank mechanism, as shown in Almandeel et al. (2015). This mechanism is applied in machines, such as driving the ram of a mechanical press. Figure 5 depicts the kinematic diagram, including the coordinates and geometrical parameters utilized for this third example. The mechanism is driven by the

variable α , and the goal is for point 3 on the coupler link to follow a predefined path (Table 6) Hernández et al. (2021). The vector of NC used in this simulation is $\{x_2, y_2, x_3, y_3, x_4, \alpha\}^T$. A total of n equal to six coordinates is used, where the orientation coordinate α is the independent coordinate and the rest are the dependent coordinates. So, in this case, the number of constraint equations m equals 5. The geometrical parameters of the mechanism are expressed by the vector $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8\}^T$, in which a_8 is the parameter that describes the slope of the straight line.

The constraint equations of this mechanism are the components of the following equation vector $\phi(q_j)$:

$$\phi(q_j) \equiv \begin{Bmatrix} (x_2 - a_6) - a_1 \cos(\alpha) \\ (y_2 - a_7) - a_1 \sin(\alpha) \\ (x_4 - x_2)^2 + (a_8 x_4 + a_5 - y_2)^2 - a_2^2 \\ (x_3 - x_2)^2 + (y_3 - y_2)^2 - a_3^2 \\ (x_4 - x_3)^2 + (a_8 x_4 + a_5 - y_3)^2 - a_4^2 \end{Bmatrix} = 0 \quad (16)$$

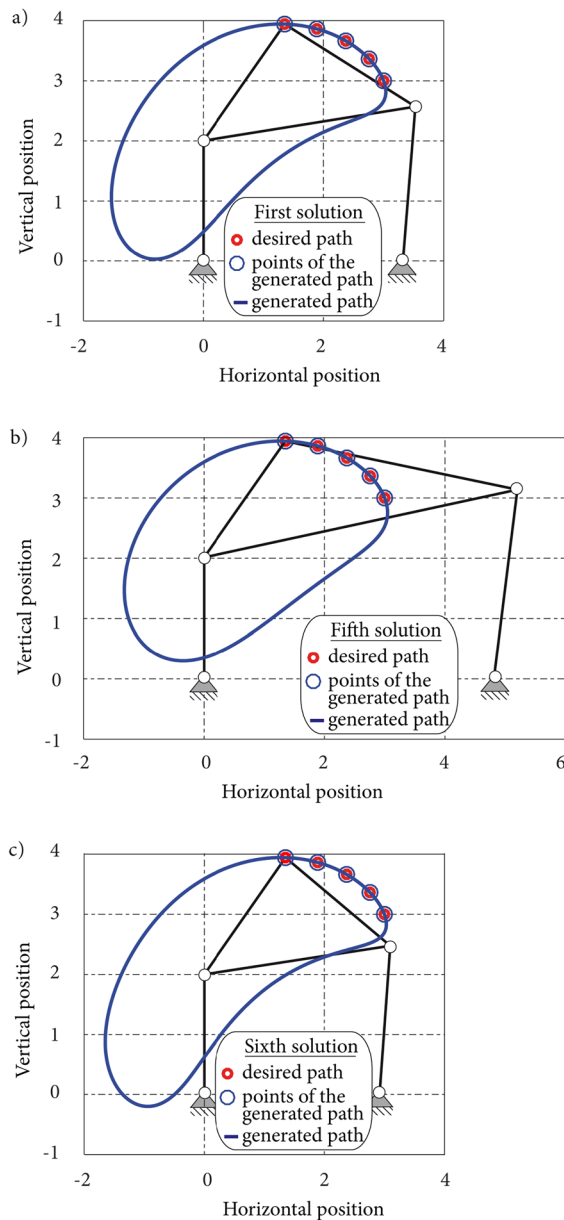


Fig. 4 Three solutions for the dimensional synthesis of a Grashof four-bar mechanism

Figure 6 illustrates the simulation of the offset slider-crank mechanism using the optimal dimensions yielded by the $NCLA_{sdm}$ method. The generated path (blue) closely approximates the desired path (red); however, a visible discrepancy remains. This deviation is a structural error inherent to the mechanism's limited degrees of freedom and geometric constraints, which prevent it from tracing the specified trajectory exactly. The algorithm has converged to the optimal structural approxi-

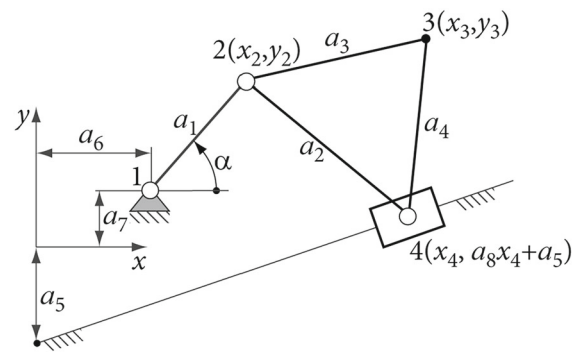


Fig. 5 NC and geometrical parameters of the offset slider-crank mechanism

Table 6 Data points of the desired path Hernández et al. (2021)

Points j	x_3	y_3	α
1	3.25	1.70	1.50
2	3.63	1.65	1.36
3	3.93	1.88	1.22
4	4.10	2.17	1.08
5	4.27	2.46	0.94
6	4.43	2.75	0.80
7	4.48	3.13	0.66
8	4.25	3.43	0.52
9	4.00	3.50	0.38
10	3.71	3.50	0.24
11	3.43	3.50	0.10
12	3.14	3.50	-0.04
13	2.86	3.50	-0.18
14	2.57	3.50	-0.32
15	2.28	3.50	-0.46
16	2.00	3.50	-0.60

mation, providing the best possible fit for this specific mechanism topology.

Table 7 shows the vector of initial IC and the error of the path, defined by the square of the Euclidean norm between the generated path and the desired path. Also, the solutions and computation time for simulations done with different numbers of iterations are shown. The simulations were performed on an HP EliteBook 830 G10 i7 computer with 32 GB of RAM.

As shown in Table 7, the parameter variation trend demonstrates rapid initial convergence followed by asymptotic stability. A significant reduction in error from 29.7 to 0.473 is achieved within the first 200 iter-

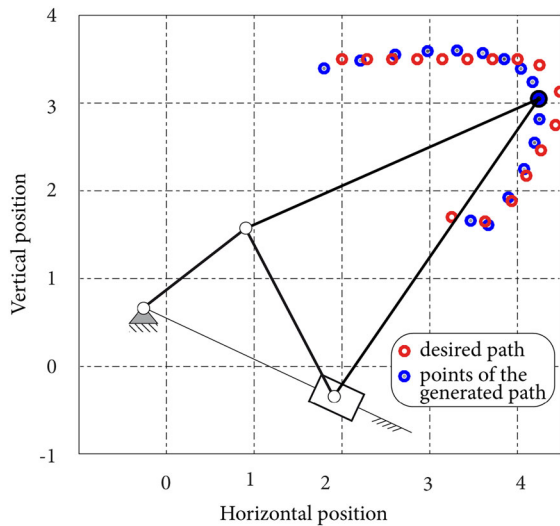


Fig. 6 Desired and generated paths for the slider-crank mechanism

ations (taking only 1.4 min). Beyond 500 iterations, the variations in parameters a_1 through a_8 become marginal, with the total error plateauing around 0.410. This indicates that the slider-crank model achieves optimal synthesis early in the process, and that further increases in iteration count (up to 60,000) yield diminishing returns in accuracy relative to the substantial increase in computation time.

Figure 7 shows convergence as a function of the iteration number. It can be seen that the algorithm achieves a high convergence rate up to iteration 50, after which convergence is very slow. Table 7 shows the dependence of some lengths a_i on iteration. This table shows that the slope a_8 of the straight line defining the slider guide is positive at iteration 2000, whereas at iteration 20,000 it is negative. Both iteration values are in the slow-convergence zone and are close to the solution. However, the values of the a_8 parameter found are sensibly different.

3.4 Example of the Theo Jansen mechanism to generate a desired path

The last example presented here is the Theo Jansen walking mechanism, a popular mechanism among legged robotics researchers. The Jansen linkage mechanism consists of 12 links connected by revolute joints and is a one *dof* mechanism Nansai et al. (2013). Figure 8 depicts the kinematic diagram and the coordinates

and geometric parameters used in this fourth example. The input of the mechanism is defined by the variable α , and it is desired that point 8 on the end-effector traces a desired path. The vector of NC used in this simulation is $\{x_2, y_2, x_3, y_3, x_4, y_4, x_6, y_6, x_7, y_7, x_8, y_8, \alpha\}^T$. A total of n equal to 13 coordinates is used, where the orientation coordinate α is the independent coordinate and the rest are the dependent coordinates. The number of constraint equations m equals 12. The geometrical parameters of this mechanism are expressed by the vector $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}, a_{11}, a_{12}, a_{13}, a_{14}, a_{15}\}^T$.

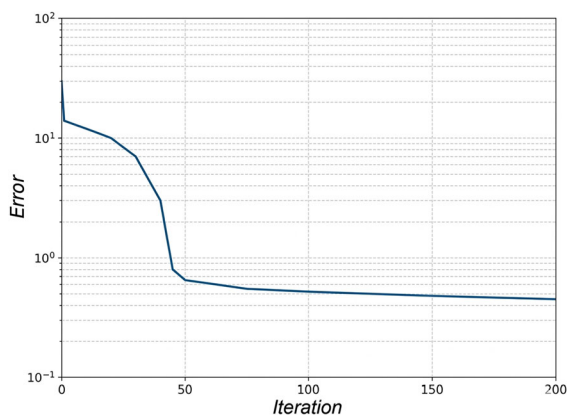
The constraint equations of this mechanism are the components of the following equation vector $\phi(q_j)$:

$$\phi(q_j) \equiv \begin{Bmatrix} (x_2 - a_{14}) - a_1 \cos(\alpha) \\ (y_2 - a_{15}) - a_1 \sin(\alpha) \\ (x_3 - x_2)^2 + (y_3 - y_2)^2 - a_2^2 \\ (x_3 - a_4)^2 + (y_3 - a_5)^2 - a_3^2 \\ (x_4 - a_4)^2 + (y_4 - a_5)^2 - a_7^2 \\ (x_6 - a_4)^2 + (y_6 - a_5)^2 - a_8^2 \\ (x_4 - x_3)^2 + (y_4 - y_3)^2 - a_6^2 \\ (x_7 - x_4)^2 + (y_7 - y_4)^2 - a_{11}^2 \\ (x_6 - x_2)^2 + (y_6 - y_2)^2 - a_9^2 \\ (x_7 - x_6)^2 + (y_7 - y_6)^2 - a_{10}^2 \\ (x_8 - x_7)^2 + (y_8 - y_7)^2 - a_{12}^2 \\ (x_8 - x_6)^2 + (y_8 - y_6)^2 - a_{13}^2 \end{Bmatrix} = 0 \quad (17)$$

Figure 9a shows Theo Jansen's walking mechanism and the desired and generated paths for maximum stride. The initial IC of the reference mechanism is taken from Jansen (2007), where the origin of the reference system is located at point 1. Nine points have been selected from the path generated from the IC of the aforementioned mechanism. These 9 points constitute the desired path of point 8 on the end-effector, that is, the point of the mechanism that makes contact with the ground when walking, to find the dimensions of the mechanisms shown in Fig. 9. We start with a known solution to this problem [(IC taken from Jansen (2007))] and generate four different strides. This is done by adding a constant value Δx to the horizontal coordinate of point 8 of the desired path. The newly four defined paths are found by adding Δx equal to $\{0, -3, 5, 10\}$ to the desired path (Fig. 9), thus obtaining maximum strides of $\{65.98, 59.00, 77.51, 88.12\}$ respectively, as can be seen in Table 8. It also shows the error defined by the Euclidean norm of the generated path relative to the desired path, and the cumulative error (Error

Table 7 Parameters a_i obtained according to different numbers of iterations and calculation time

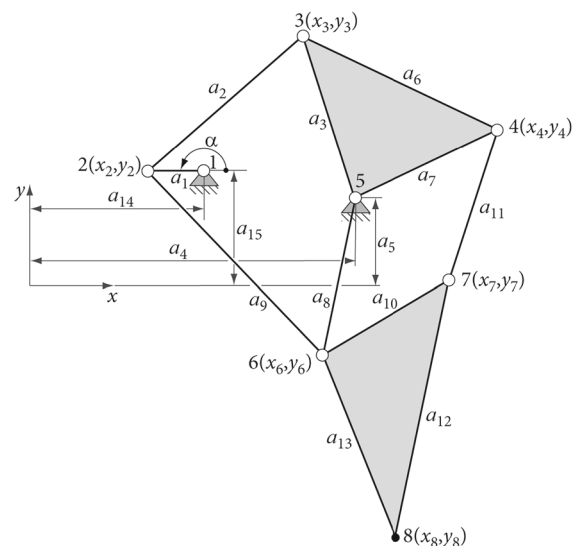
Iterations ($h_0 = 0.001$)			200	500	2000	20000	60000
Time (min)		IC	1.4	3.7	14.2	160.3	502.0
Parameters a_i	a_1	1.0	1.4649	1.4800	1.4822	1.4399	1.4692
	a_2	3.0	2.0414	2.0127	2.0271	2.0562	2.1625
	a_3	3.0	3.7144	3.7237	3.7262	3.5527	3.6520
	a_4	3.0	2.8466	2.8744	2.9769	3.9612	4.1192
	a_5	0.5	0.2014	0.2403	0.2126	0.0926	− 0.0652
	a_6	0.0	− 0.3615	− 0.3325	− 0.3374	− 0.1489	− 0.2593
	a_7	0.0	0.6066	0.6230	0.6233	0.7347	0.6704
	a_8	− 0.3	0.1719	0.6230	0.1300	− 0.1457	− 0.1466
Error	29.7	0.473	0.467	0.466	0.413	0.410	

**Fig. 7** Convergence according to the number of iterations

Y) in predicting the vertical position (y_8). The latter error estimator specifies when the iteration stops. This is intended to extend the maximum stride without major changes in height or vertical displacement of point 8.

4 Discussion

This section discusses the advantages and limitations of the proposed dimensional synthesis approach. Example 3.1 shows how the method works on an inline slider-crank mechanism. As shown, the solution to this problem can be found in two steps. Also, it is possible to find an analytical solution of the step size that maximizes the reduction along the search direction, Eq. (14), due to the simplicity of the proposed problem. In the other three examples proposed, it is no longer possible to find an analytical solution; Eq. (9) must be solved numerically.

**Fig. 8** NC and geometrical parameters of the Jansen linkage

ically. For this purpose, the Newton–Raphson method was implemented and validated, starting with a step size h_0 , as shown in Tables 4, 5, 7, and 8.

Example 3.2 shows that the dimensional synthesis method $NCLA_{sdm}$ proposed here performs effectively. Table 4 shows that the error obtained by means of the $NCLA_{sdm}$ method with 8000 iterations is of the same order of magnitude as the error obtained by the IOA^{s-ot} algorithm Ortiz et al. (2013). The main difference is that $NCLA_{sdm}$ is based on a very different methodology from the K-K and IOA^{s-ot} methods. $NCLA_{sdm}$ uses a formulation based on constraint equations, which are widely used in mechanism kinematics problems. Kinematic constraint equations are

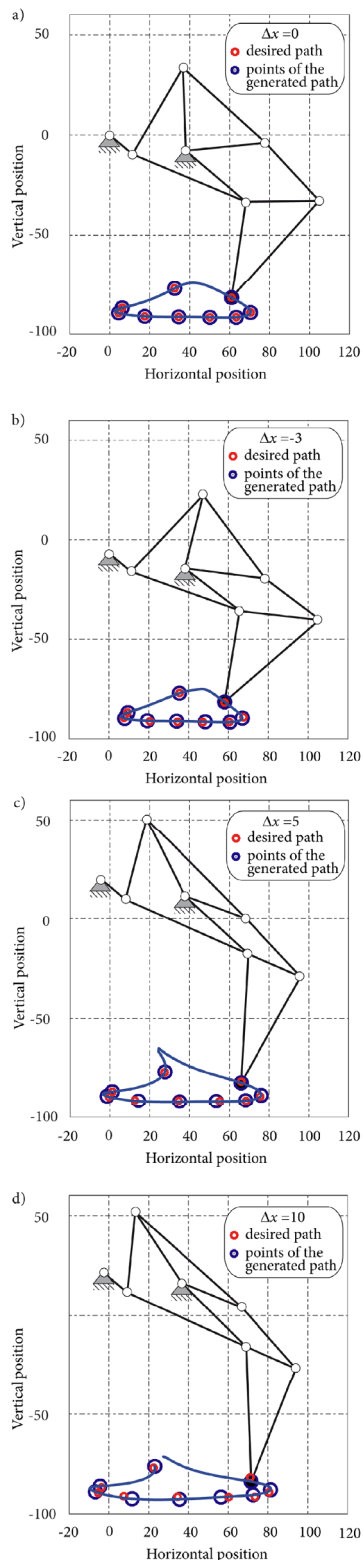


Fig. 9 Jansen linkage synthesis solutions

commonly used for modeling mechanisms and can now also be used for mechanism synthesis. This represents a significant advantage, as the same formulation used for modeling can also be used for synthesis, as shown in Eq. (15). In this example, the convergence rate in the first iterations is high, whereas it is lower near the solution. This is a known characteristic of the steepest descent method Bartholomew-Biggs (2008). The convergence of this family of methods is of first order, and as it approaches the solution, it becomes very slow. In future work, we will attempt to implement a second-order method to accelerate convergence.

On the other hand, Table 5 shows how this type of problem, defined with the above-mentioned constraints, has multiple possible solutions. Starting from different ICs, different solutions, geometries, and generated paths are obtained, except at the desired points. Figure 4 shows how different initial conditions minimize the error between the path generated by the mechanism and the desired path. All the solutions found have an error of the same order of magnitude and are therefore comparable (Table 5). The articles that develop the K-K and IOA^{s-ot} methods only present a single optimal solution, suggesting that there is only one four-bar linkage that minimizes the error. This article shows that there is more than one four-bar mechanism that solves the kinematic problem; thus, this represents a significant advantage over methods reported by other authors.

The four-bar mechanism is extensively studied, and numerous studies focus on its synthesis: Avilés et al. (2010) employs gradient-based optimization for path synthesis, Almandeel et al. (2015) utilizes an evolutionary algorithm-based optimization, and Cabrera et al. (2011) applies a modified Krill Herd algorithm for this type of mechanism's synthesis. All these studies employ different optimization methods and show the mechanism's path passing through a set of desired points. However, the specialized bibliography usually does not address the possibility of finding more than one solution for a given desired path. There is also no reference to additional constraints to define, for example, the length or smoothness of the desired path, which could lead to a unique solution.

In example 3.3, the synthesis method is applied to an offset slider-crank mechanism, which is a problem with 8 dimensional parameters to be determined. There is a high convergence rate up to iteration 50; thereafter, it decreases sharply as the solution approaches the opti-

Table 8 Parameters a_i obtained according to different Δx of point 8 in a Jansen's walking mechanism

Modified stride values		Step size $h_0 = 0.001$ IC Jansen (2007)	$\Delta x = 0$	$\Delta x = -3$	$\Delta x = 5$	$\Delta x = 10$
Parameters a_i	a_1	15.0	15.09	14.67	15.87	15.94
	a_2	50.0	50.03	52.85	42.24	40.43
	a_3	41.50	41.47	38.05	43.58	42.36
	a_4	38.0	38.09	38.09	37.69	36.46
	a_5	- 7.80	- 7.80	- 14.92	11.38	16.72
	a_6	55.80	55.82	52.16	71.15	72.20
	a_7	40.10	40.09	40.78	32.86	33.10
	a_8	39.30	39.43	34.60	43.14	45.78
	a_9	61.90	61.53	57.50	67.46	65.68
	a_{10}	36.70	36.70	40.50	28.63	27.22
	a_{11}	39.40	39.46	34.06	39.86	40.69
	a_{12}	49.00	48.90	46.15	64.71	68.41
	a_{13}	65.70	65.70	62.98	61.06	61.57
	a_{14}	0.00	- 0.15	0.15	4.32	- 3.00
	a_{15}	0.00	0.16	- 7.21	19.75	21.99
Error		29.0	0.04	4.35	8.74	47.90
Error Y (y_8)		16.89	0.02	0.5	1.21	11.65
Stride			65.98	59.00	77.51	88.12

mum. On the other hand, the slope a_8 of the slider guide changes sign as the number of iterations increases from 2000 to 20,000 (Table 7). This change is significant because the problem is sensitive to the guide's slope. The parameter a_8 converges to the solution of the problem in the low-convergence-rate zone, which justifies the implementation of quadratic-convergence methods. For this example, the simulation time cost to reach 60000 iterations is comparatively high compared to the previous examples.

In example 3.4, it is shown that Theo Jansen's walking mechanism has 15 parameters to be determined. This is a complex problem, so we start with the previous solution found for Theo Jansen's linkage and perturb the desired path by applying a Δx to point 8 on the end-effector to achieve different strides. The algorithm proposed in this work allows us to determine the mechanism's dimensions for different strides, which in this case range from 59.0 to 88.1 (Table 8), with minimal variation in the vertical component (y_8) during the stepping action.

While the primary objective of this work is to establish a synthesis framework based on kinematic constraint equations rather than to maximize numerical

convergence, several avenues for algorithmic enhancement remain. Although the classical Steepest Descent Method (SDM) is known for its slow convergence when approaching the optimum, primarily due to its fixed-step progression and oscillatory trajectory, the proposed $NCLA_{sdm}$ formulation mitigates these limitations through its dual-mechanism structure, including the determination of the optimum step, thereby enhancing directional stability and convergence. The adaptive Lagrangian components act as a dynamic regularization term that locally convexifies the objective surface and improves numerical conditioning. As a result, the descent direction becomes more stable and efficient, allowing the algorithm to maintain a consistent convergence rate even in the high-accuracy region (with a tolerance of 10^{-6}). Qualitatively, this behavior aligns with the faster convergence traits typically associated with advanced gradient-based schemes such as the conjugate gradient or quasi-Newton methods, while retaining the conceptual simplicity of the steepest descent framework. The current use of the steepest descent method provides a robust, easily implemented foundation, but its relatively slow convergence suggests that integrating second-order methods is a logical next step. Further-

more, while the introduction of random perturbations could improve search capabilities, our observation of multiple valid solutions (see Fig. 4) indicates that such stochastic elements must be applied cautiously to avoid undesirable jumps between distinct mechanism topologies.

Beyond these numerical refinements, the structural scope of the $NCLA_{sdm}$ framework is also poised for expansion. Although this study focuses on planar mechanisms, the method is mathematically compatible with spatial trajectories. Extending the approach to 3D systems would simply involve redefining the natural coordinate vector q to include the NC and unit vectors in three-dimensional space, and updating the kinematic constraint equations accordingly. Validating the proposed algorithm for the dimensional synthesis of spatial linkages, targeting complex systems such as the RSSR linkage or the Stewart platform, represents a promising research direction that builds upon the foundational efficiency demonstrated in this planar study.

5 Conclusions

This work proposes a method for the planar mechanism dimensional synthesis problem using natural coordinates in a Lagrangian optimization algorithm. The main conclusions of this study are as follows:

- A method has been developed and implemented for performing the dimensional synthesis of bar mechanisms from constraint equations in natural coordinates. The core innovation of this $NCLA_{sdm}$ approach lies in its algorithmic simplicity; by using a first-order framework, it achieves high-precision convergence while bypassing the matrix instabilities and second-order derivative requirements typical of more complex solvers.
- The proposed method has been shown to converge to a solution in several mechanisms, thereby generating a configuration or solution to the synthesis problem. Synthesis solutions have been generated for mechanisms with 2, 6, 8, and 15 geometric parameters. The method demonstrated high stability across different initial conditions, consistently reaching an error floor in the range of 10^{-5} to 10^{-6} , thereby proving its robustness to the initial-value problem common in kinematic synthesis.
- It has been proven that the Steepest Descent Method slows down the convergence when it is close to

the solution. However, the results show that the Lagrangian component serves as a dynamic regularization term, stabilizing the descent direction. This allows the algorithm to maintain competitive performance, achieving optimal synthesis for complex linkages within a reasonable timeframe of 62.9 min.

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Author contributions HAGR and AJSE proposed the idea of the method and paper; EEZF, AENA and EDCP state of the art; HAGR and EEZF performed the numerical simulation and presented the numerical details; HAGR, EEZF and AJSE analyzed the results; HAGR and AJSE validation and testing of the method; AJSE, AENA and EDCP proofread, revised, and structured the article. All authors read and approved the final manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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